A

PROJECT REPORT

ON

“Student Career Prediction System using Machine Learning”

Submitted to

MSBTE

In partial fulfillment of degree of

DIPLOMA COMPUTER SCIENCE AND ENGINEERING

Submitted by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

TATYASAHEB KORE INSTITUTE OF ENGINEERING & TECHNOLOGY (POLYTECHNIC)

Warananagar , Dist. Kolhapur (2024-25)

TATYASAHEB KORE INSTITUTE OF ENGINEERING & TECHNOLOGY (POLYTECHNIC)WARANANAGAR,

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CERTIFICATE

This is to certify that the project report entitled

"Student Career Prediction System using Machine Learning"

Is a bonafide work of Shreya.U.Bachche , Apeksha.M.Bhosale , Vaishnavi.U.Chougule, Samiksha.A.Patil, Gayatri.S.Sutar in **the partial fulfillment of the requirements for the award of diploma of engineering in Computer Science and Engineering of the MSBTE. They have carried out the work under my supervision and guidance, during academic year 2024-2025.**

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Prof.P.R.Patil
Principal

DECLARATION

We hereby declare that the project report entitled **““Student Career Prediction System using Machine Learning””** has been prepared by us under guidance and supervision of guide **Prof.S.D.Jadhav** TATYASAHEB KORE INSTITUTE OF ENGINEERING & TECHNOLOGY (POLYTECHNIC), Warananagar, Dist. Kolhapur in partial fulfillment of the degree of “DIPLOMA IN COMPUTER SCIENCE AND ENGINEERING” is our original work and not directly copied from any source.

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Place: Warananagar (Kolhapur)

Your Sincerely,

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Samiksha Anil Patil

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ACKNOWLEDGEMENT

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ABSTRACT

The Student Career Prediction System using Machine Learning with the Random Forest algorithm is developed to guide students toward suitable career paths by analysing their performance in a Multiple Choice Question (MCQ) based assessment. The system evaluates students' strengths, interests, and aptitude across various domains through the exam and uses the results as input features for the Random Forest classifier. This ensemble learning method processes the data to predict the most appropriate career options with high accuracy and reliability. By combining automated assessment with intelligent prediction, the system provides personalized career guidance, helping students make informed and confident decisions about their future.

INTRODUCTION

A Career Prediction System for Students, powered by Machine Learning, is designed to assist students in making well-informed decisions about their career paths as they approach the end of their school years, particularly after completing the 10th and 12th grades. At these critical junctures, many students face uncertainty in choosing a career direction, whether it's selecting a stream such as science, commerce, or arts, or deciding on a specific profession like engineering, medicine, business, or design. This system leverages machine learning techniques to evaluate a student's academic performance, interests, strengths, and personal preferences, and then recommends potential career options that align with their profile.

LITERATURE REVIEWS

Sr. No.	Name of an Author	Name of Paper	Year	Tools	Abstract
1.	Vidyapriya.C,Vishhnuvardhan.R.C	STUDENT CAREER PREDICTION	2017	Machine Learning and It's Algorithm	As students are going through their academics and pursuing their interested courses, it is very important for them to assess their capabilities.
2.	Prathamesh Gavhane[1], Dhanraj Shinde[1], Ashwini Lomte[1], Naveen Nattuva[1], Monika Munjal[1]	Career Path Prediction Using Machine Learning	2020	Machine Learning, Classification algorithms	In today's era, choosing the right career option is the challenging task. Starting at the early stage of life students usually fail to grasp the idea of which career to pursue as they lack maturity and the experience related to that field.

PROBLEM STATEMENT

The problem is that students lack the tools to make informed decisions about their careers. They often rely on subjective factors like family advice, peer pressure, or incomplete information about the job market and the required skill sets for different professions. This leads to a mismatch between the chosen career path and the student's true potential, which can result in dissatisfaction or wasted years in pursuing a field that may not be the right fit.

1 . Challenges in Career Decision-Making: Students struggle to make informed career choices in a dynamic and competitive environment.

2 . Lack of Guidance and Awareness: Many students face difficulties due to a lack of proper guidance, self-awareness, and understanding of market demands.

3 . Role of a Career Prediction System: The system leverages data-driven techniques to analyze students' skills, interests, academic performance, and personal traits.

4 . Use of Machine Learning: Machine learning models analyze data to predict the most suitable career paths for students.

SOFTWARE REQUIREMENTS

- Operating System: Windows 10
- Integrated Development Environment (IDE): Visual Studio Code.
- XAMPP Server: Latest stable version that is v3.3.30.
- MySQL Database: Latest stable version or compatible relational database management system (RDBMS), port 3306.
- Web Browser: Google Chrome

HARDWARE REQUIREMENTS

- Processor: i3.
- RAM: 8GB
- Storage: 476GB.

ARCHITECTURE

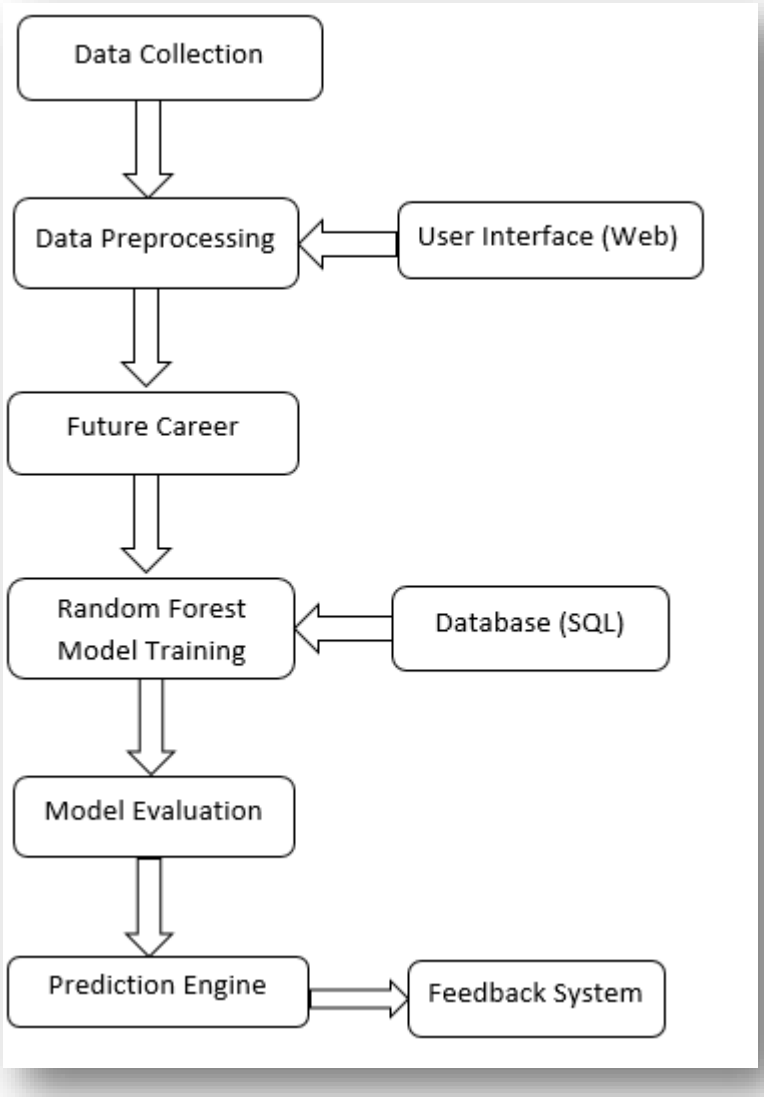


Fig1. Architecture

DATAFLOW DIAGRAM

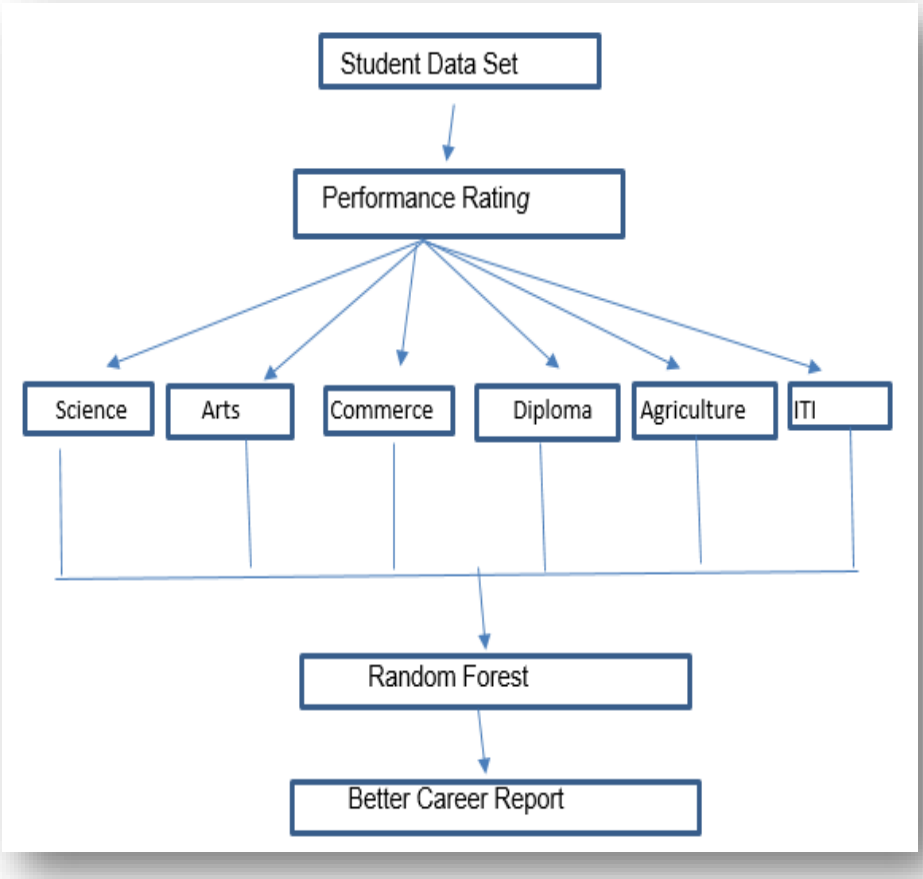


Fig2. Dataflow Diagram

SYSTEM IMPLEMENTATION

- **User Registration/Login:**
  - The user (student) registers or logs into the system using email/password credentials.
  - The user is authenticated, and their data is retrieved or stored in the database.
- **Data Collection:**
  - The student fills out a form providing their academic background, interests, hobbies, and skills.
  - This data is sent to the backend where it is processed.
- **Data Preprocessing:**
  - The collected data is preprocessed (missing values handled, categorical data encoded, and numerical features scaled).
  - This prepared data is then fed into the trained machine learning model.
- **Career Prediction via Random Forest:**
  - The **Random Forest algorithm** analyzes the input data using pre-trained decision trees to provide career predictions.
  - It generates a list of potential career paths based on the student's profile.
- **Result Display:**
  - The system displays the recommended careers with descriptions, required qualifications, and job prospects.
- **Admin Management:**
  - Admins can upload new data or modify existing datasets to improve model accuracy.
  - They can view usage statistics, feedback, and report generation.

MODULES

1. User Registration Module

- **Purpose:** To allow new students to sign up and create an account.
- **Features:**
  - Input fields: Name, email, password, age, education level, etc.
  - Data validation (e.g., unique email, password strength).
  - Stores information securely in the database.

2. Login & Authentication Module

- **Purpose:** To allow registered users (students/admins) to securely log in.
- **Features:**
  - Username and password authentication.
  - Role-based login (admin or student).
  - Password encryption.
  - Forgot Password/Reset functionality.

3. Admin Module

- **Purpose:** To allow administrators or counsellors to manage users and monitor app activity.
- **Features:**
  - View and manage student records.
  - Upload or update datasets for training.
  - Monitor prediction history and feedback.
  - Generate reports and analytics.

4. User Module (Student Panel)

- **Purpose:** To allow students to interact with the app and receive career suggestions.
- **Features:**
  - View personal profile and edit info.
  - Take career prediction exam/test.
  - See recommended careers with explanation.
  - Provide feedback on predictions.

5. Exam/Test Module

- **Purpose:** To collect data points for prediction such as:
  - Subject interest
  - Logical reasoning
  - Personality traits
  - Academic background
- **Features:**
  - Multiple-choice or rating-based questions.
  - Categorized by area: technical, creative, management, etc.
  - Auto-submit and scoring.
  - Stores answers for prediction use.

6. Career Prediction Module (ML Core)

- **Purpose:** Uses the Random Forest algorithm to predict suitable careers.
- **Features:**
  - Takes input from student answers.
  - Runs the trained Random Forest model.
  - Outputs top career matches with confidence scores.

7. Recommendation Module

- **Purpose:** Provides a list of suggested careers based on the prediction.
- **Features:**
  - Shows top 3–5 career options.
  - Includes descriptions, required skills, average salary, and demand.
  - Optional links to resources or further reading.

8. Feedback & Learning Module

- **Purpose:** Collects feedback from students to improve system accuracy.
- **Features:**
  - Feedback form after prediction.
  - Students can rate prediction accuracy.
  - Data stored for model retraining (future version).

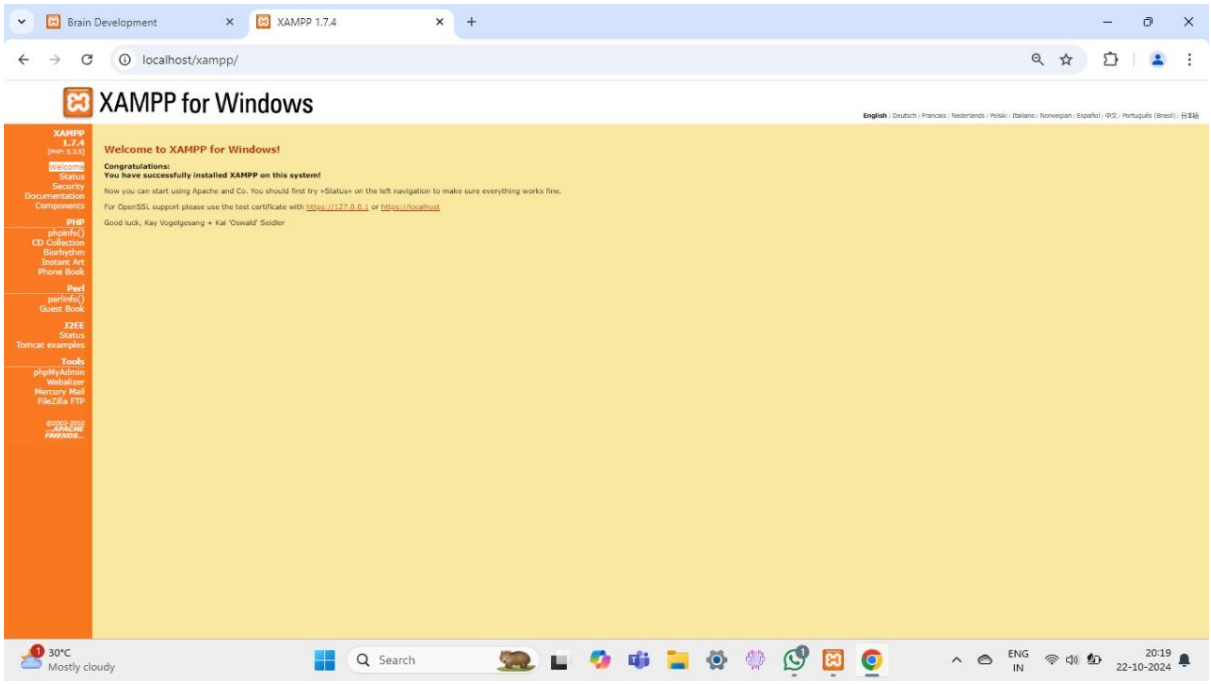
9. Database Module

- **Purpose:** Stores all system data securely.
- **Data Stored:**
  - User credentials and profiles.
  - Exam responses.
  - Prediction results.
  - Admin logs and dataset uploads.

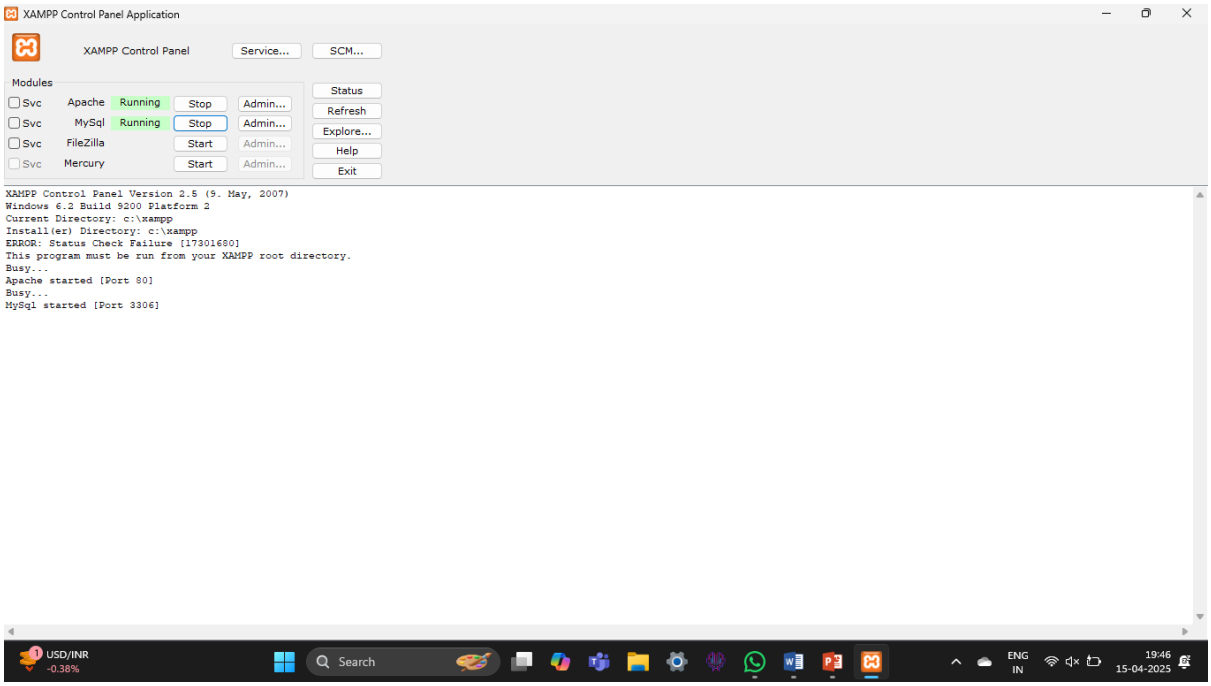
10. Notification Module (Optional)

- **Purpose:** Sends updates or suggestions to students.
- **Examples:**
  - New test available.
  - Retake prediction after 6 months.
  - New career added to database.

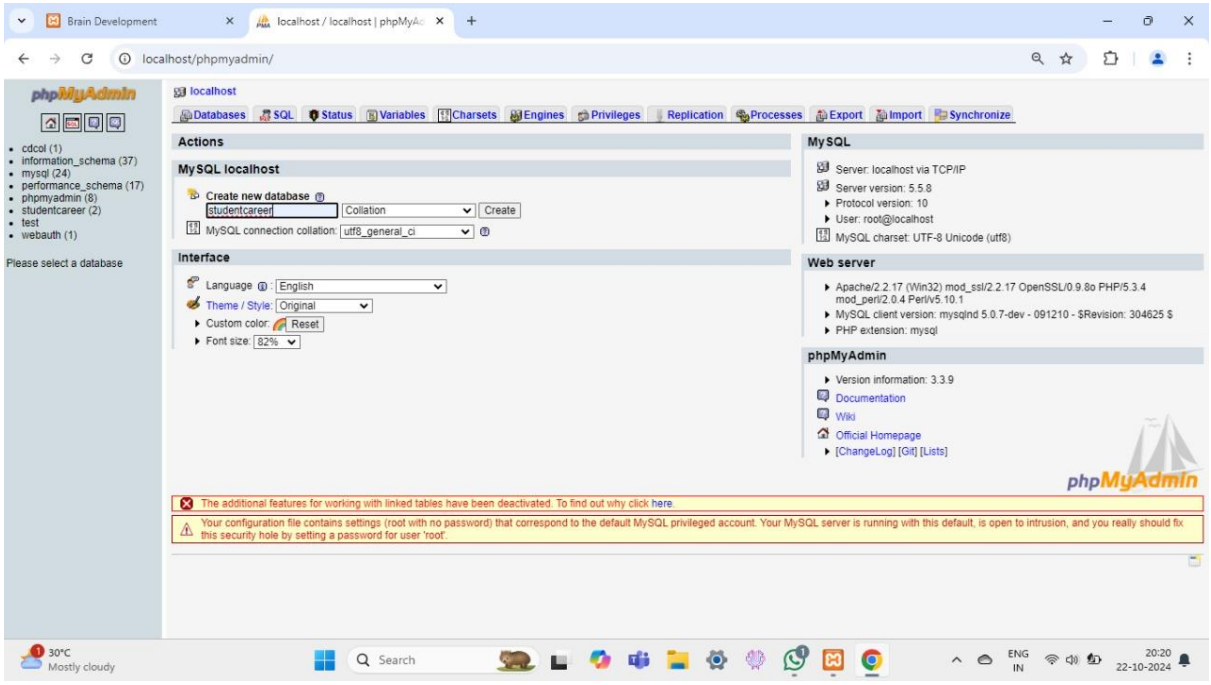
SNAPSHOT



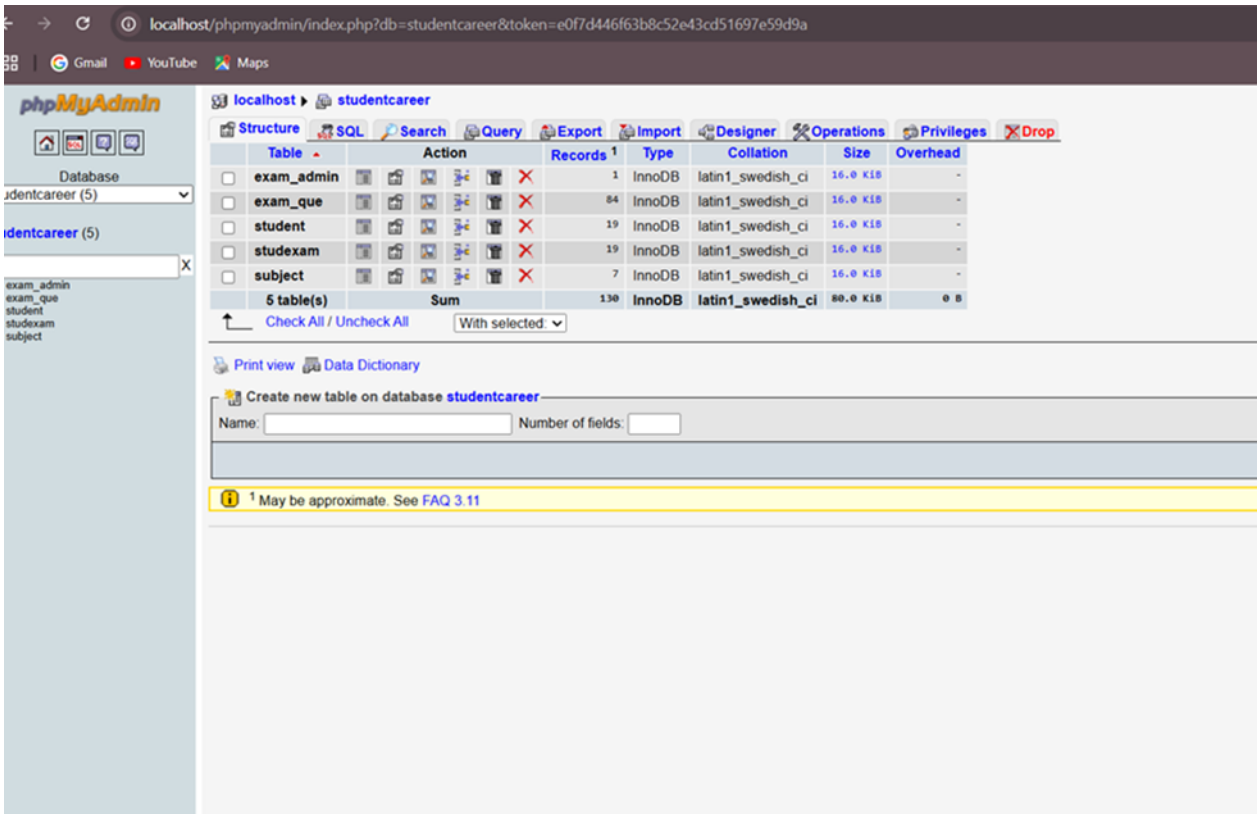
SNAPSHOT1. XAMPP Server



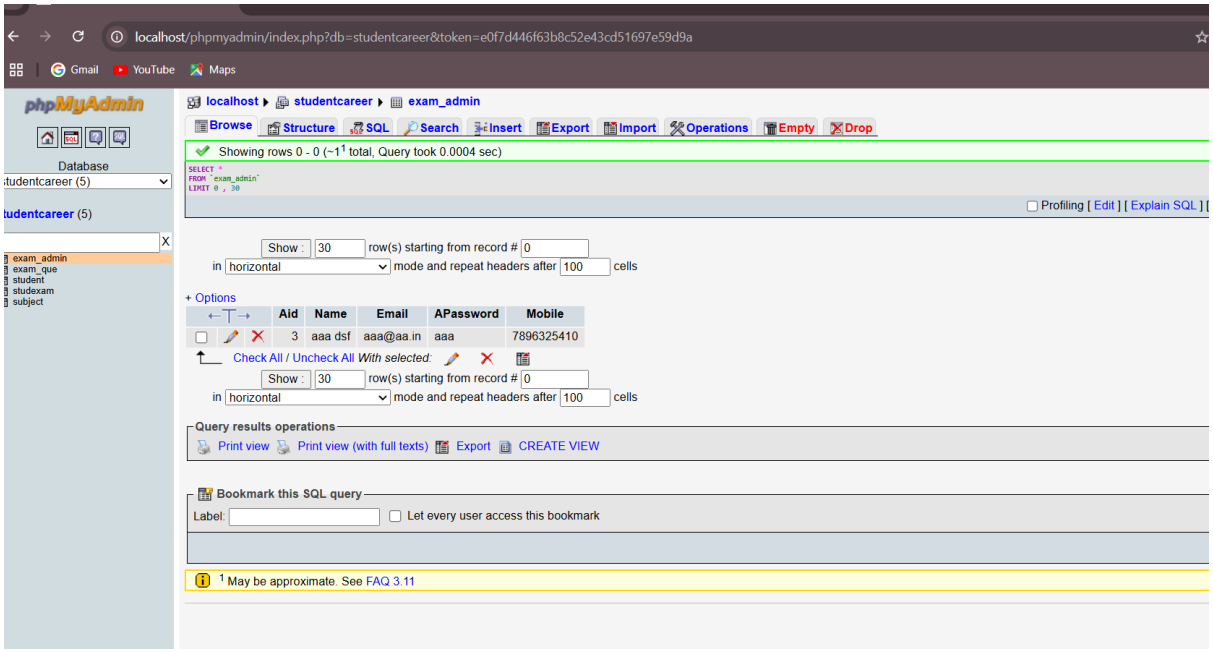
SNAPSHOT2. XAMPP Server Running



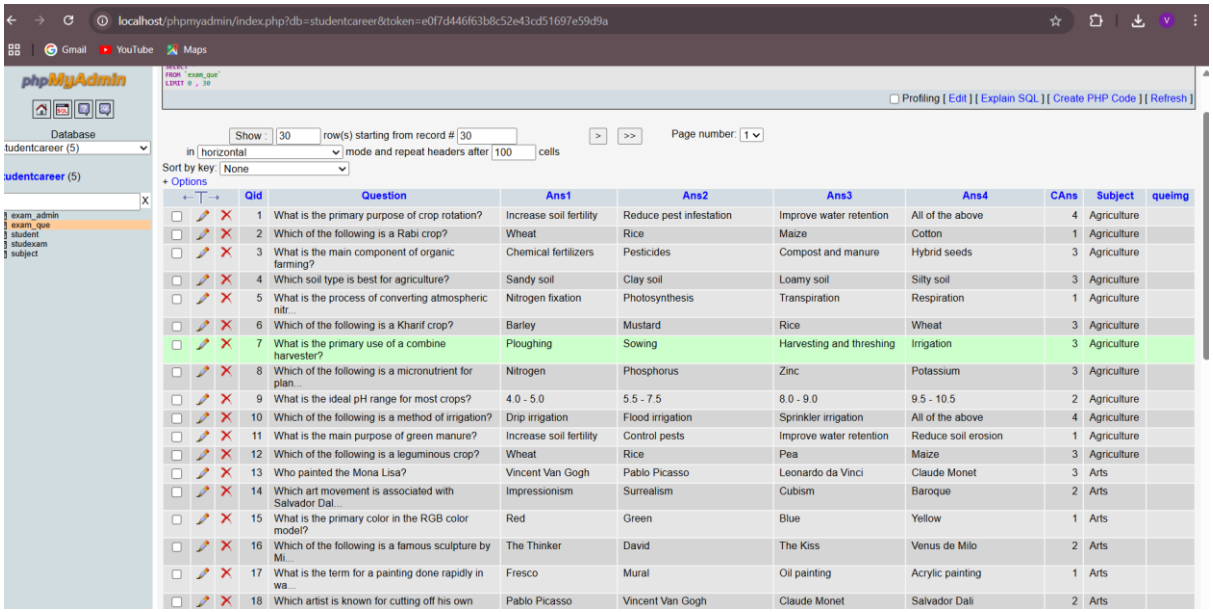
SNAPSHOT3. Database



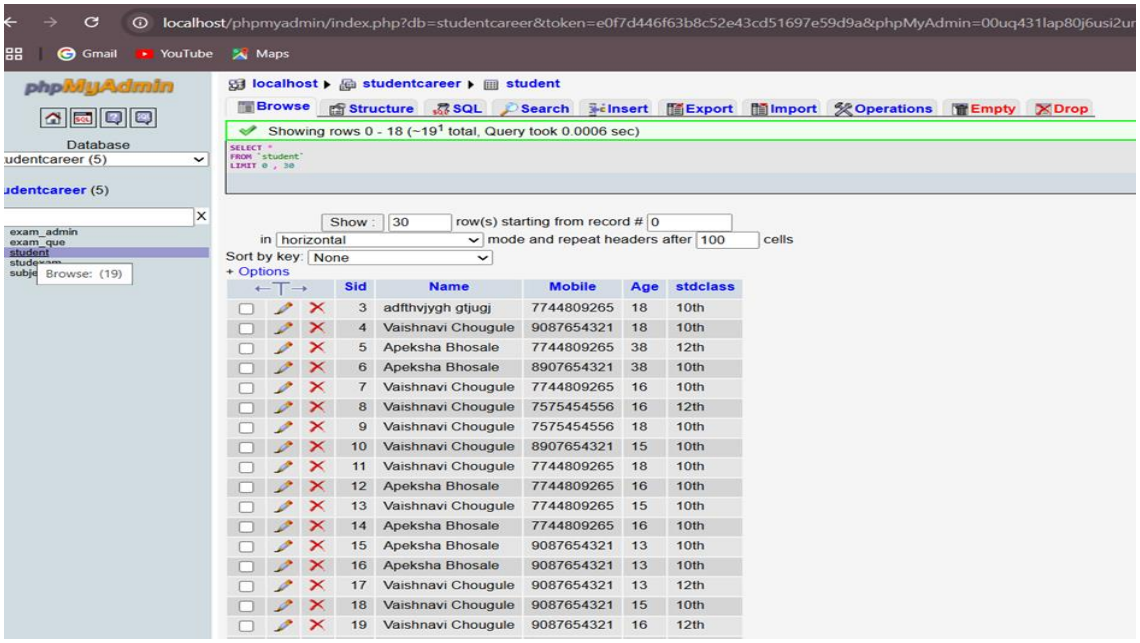
SNAPSHOT4. Create a Database



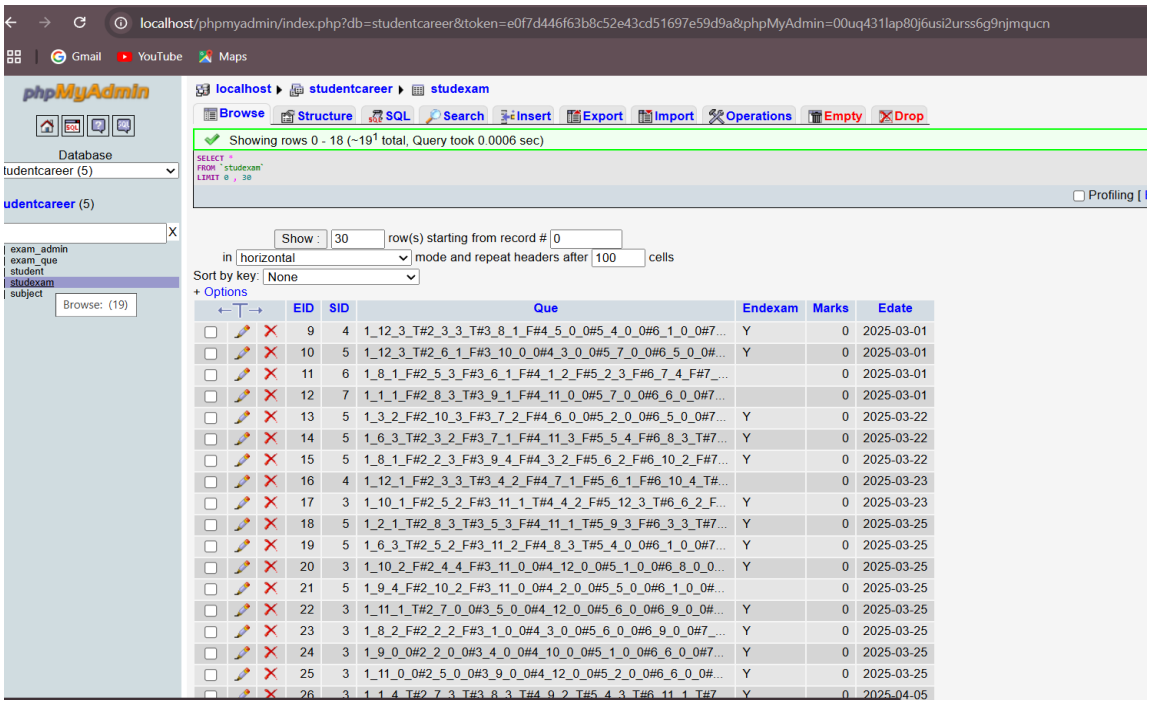
SNAPSHOT5. Admin Database



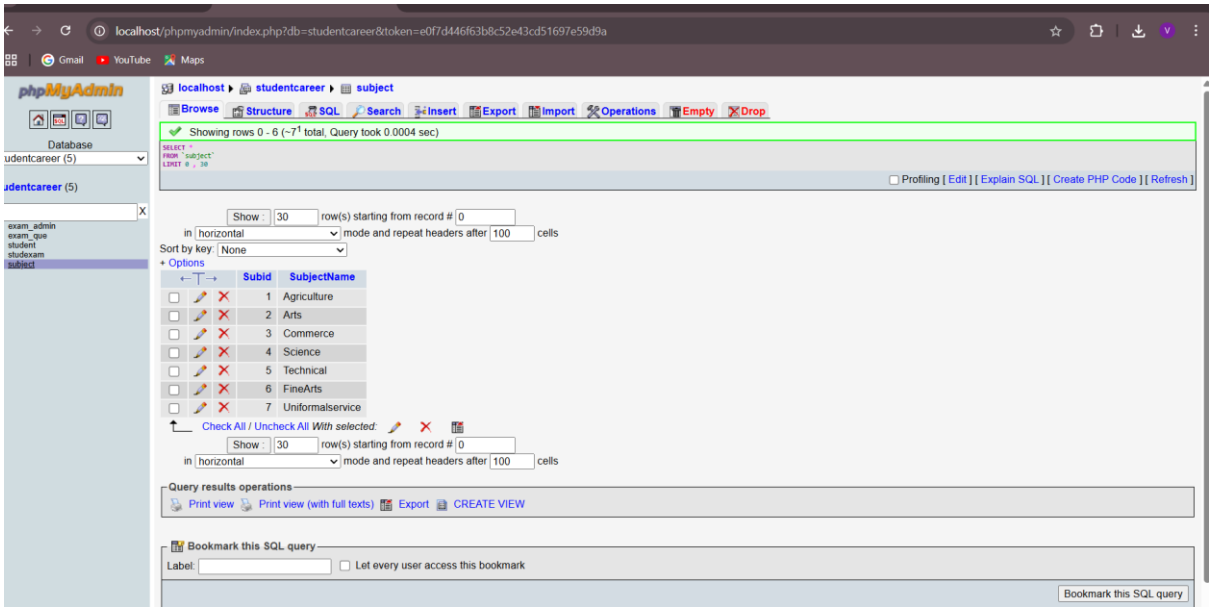
SNAPSHOT6. Question Database



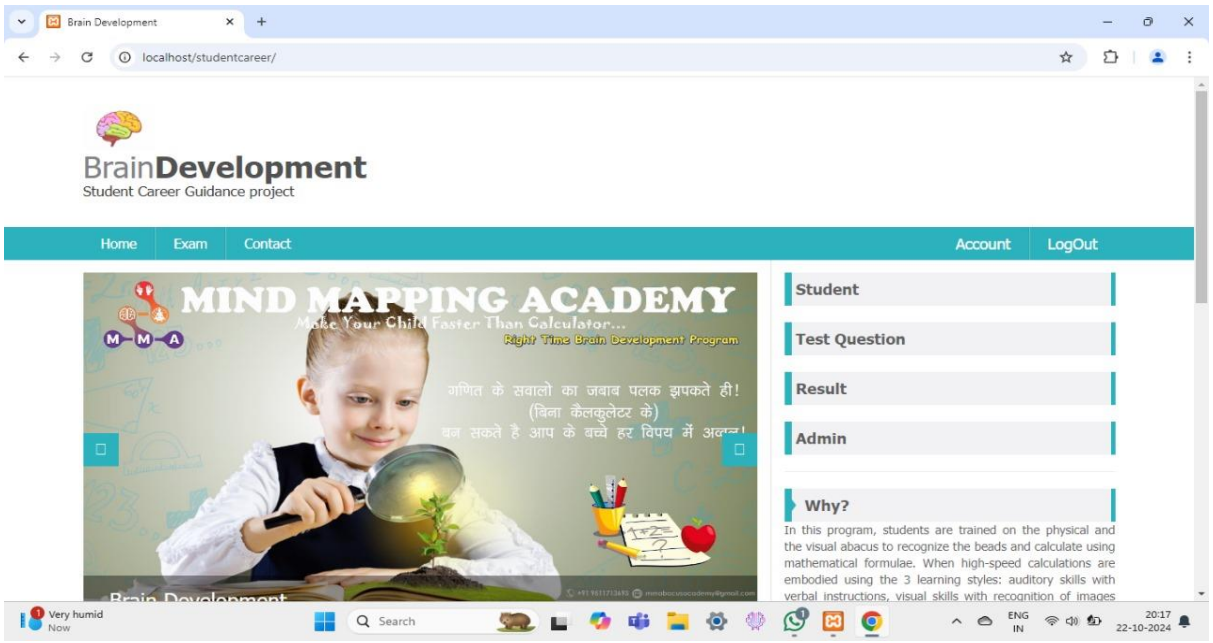
SNAPSHOT7. Student Database



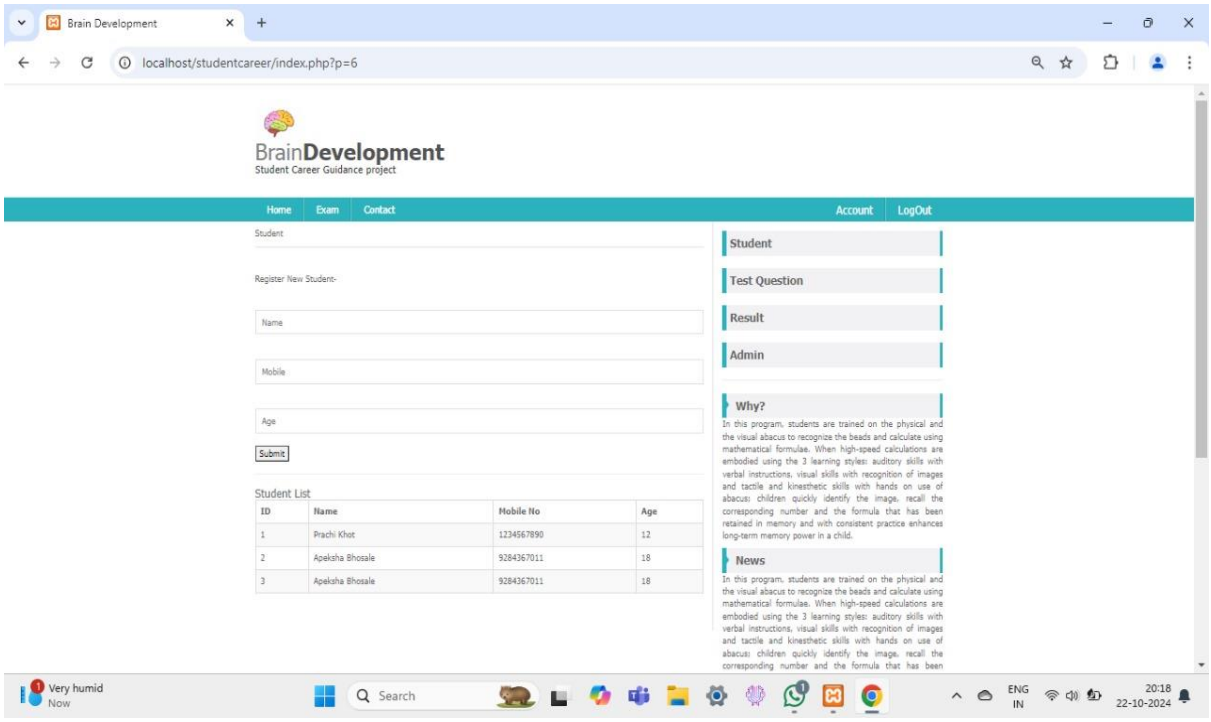
SNAPSHOT8. Exam Database



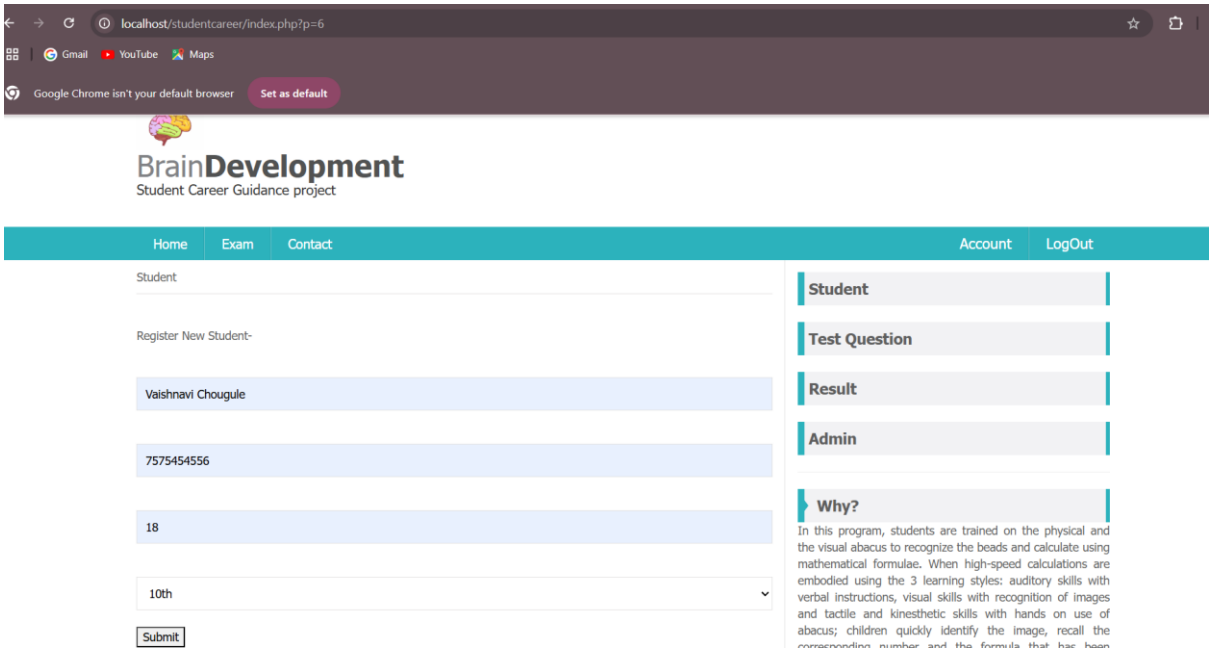
SNAPSHOT9. Career Related Fields



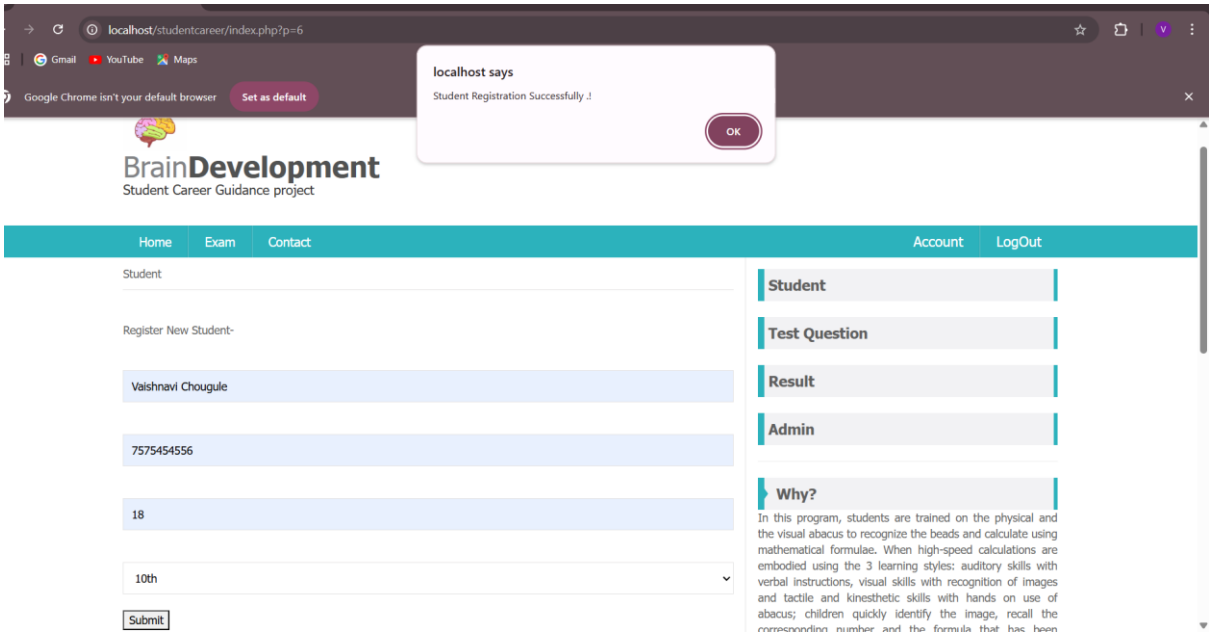
SNAPSHOT10. Login Page



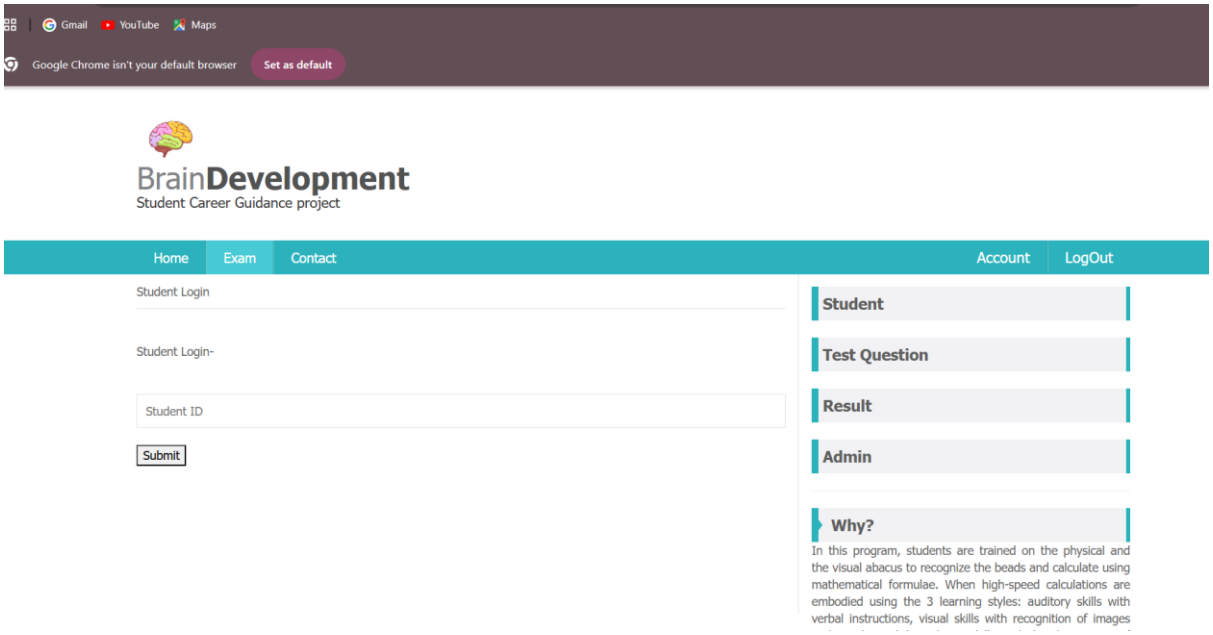
SNAPSHOT11. Student Registration



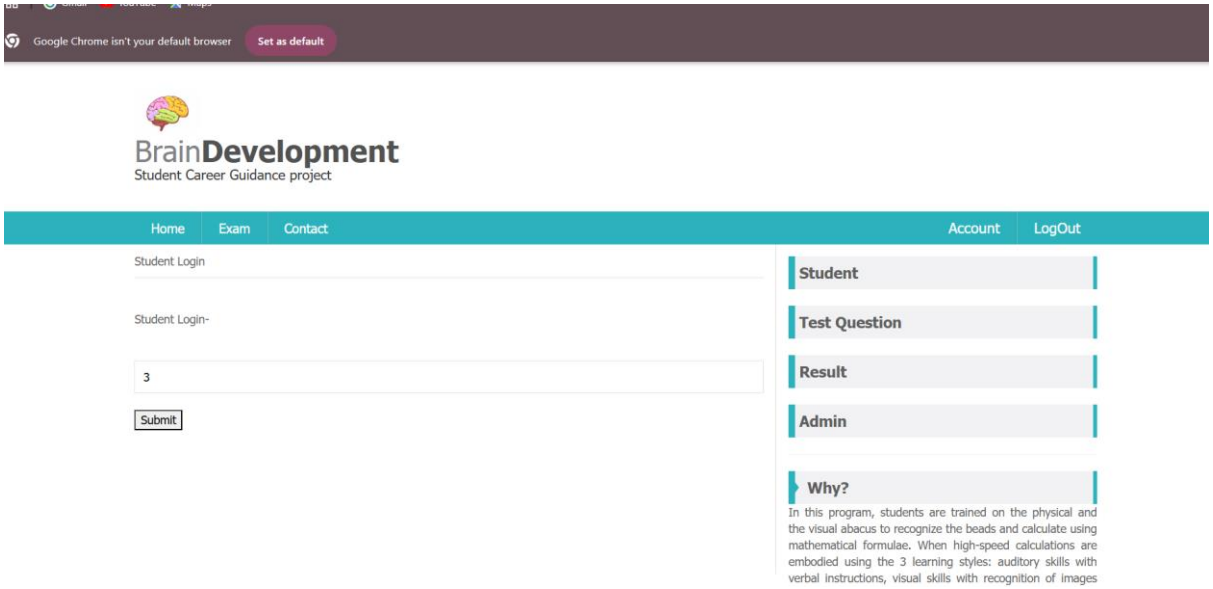
SNAPSHOT12. Student Registration Completed



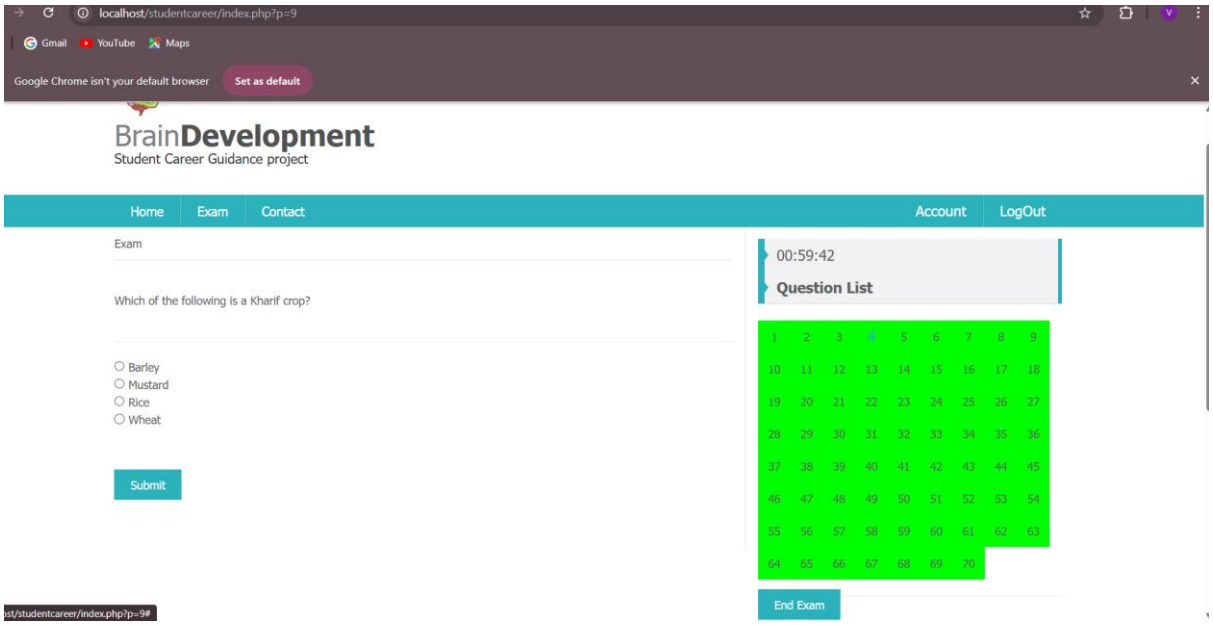
SNAPSHOT13. Student Registration Completed Successfully



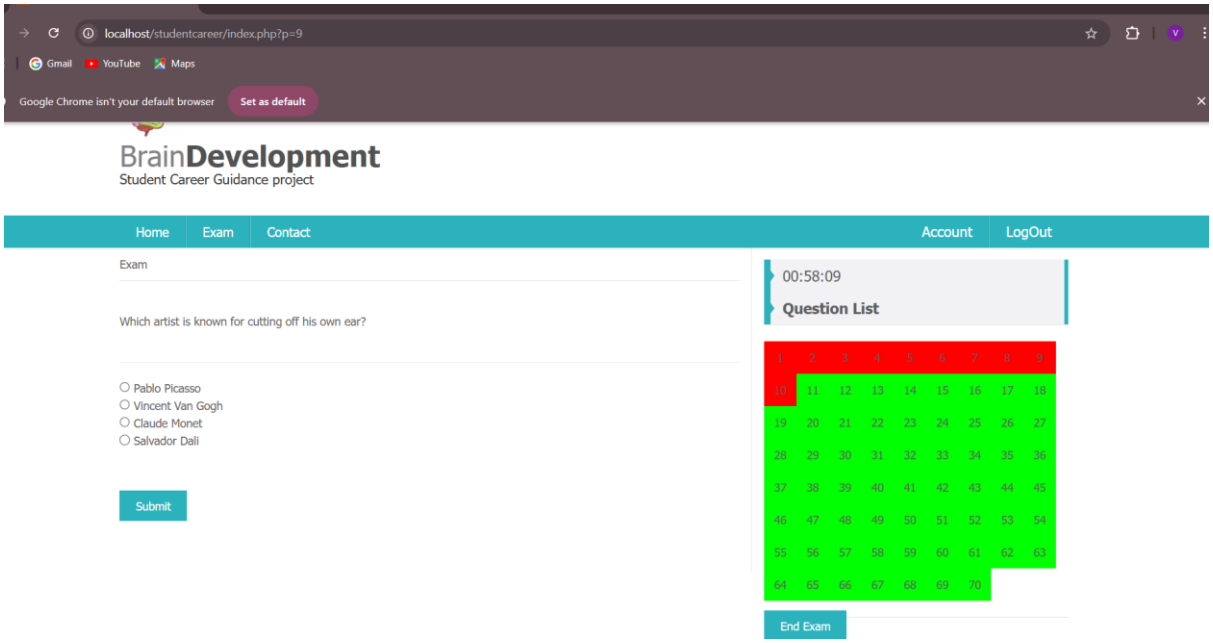
SNAPSHOT14. Student Login



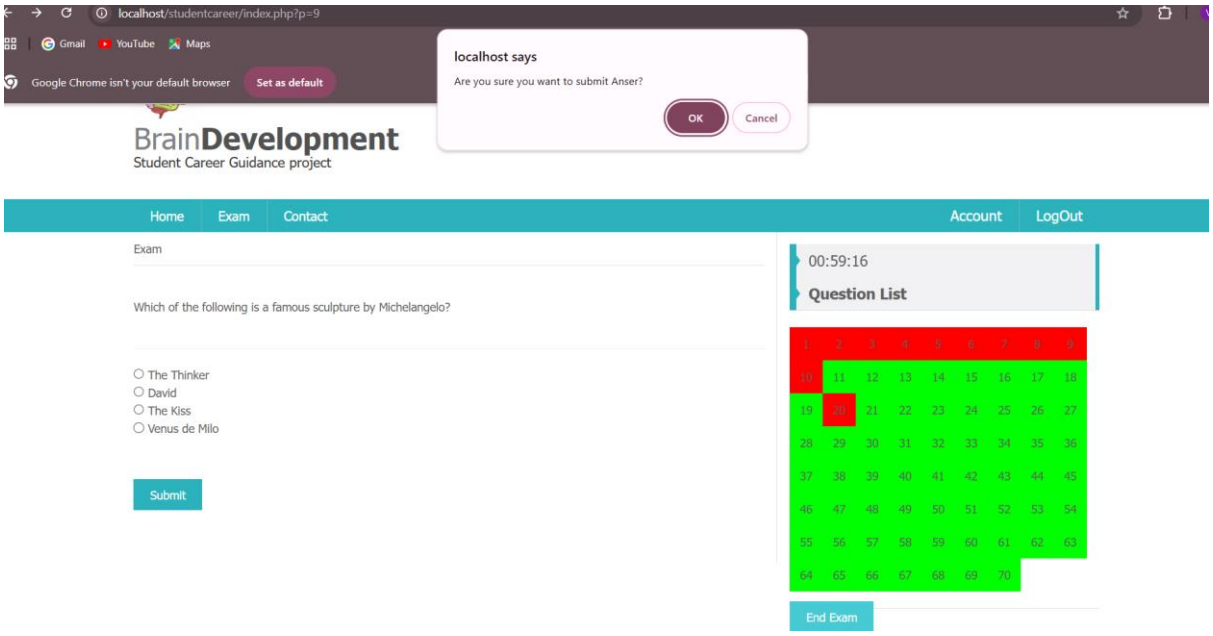
SNAPSHOT15. Login by Student ID



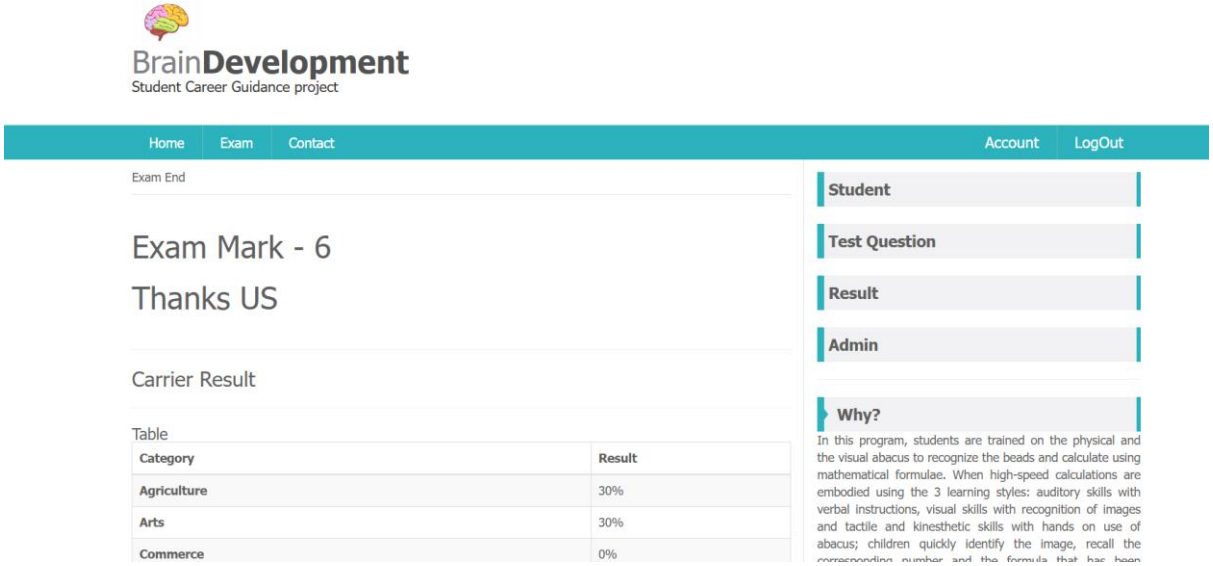
SNAPSHOT16. Student Exam Start



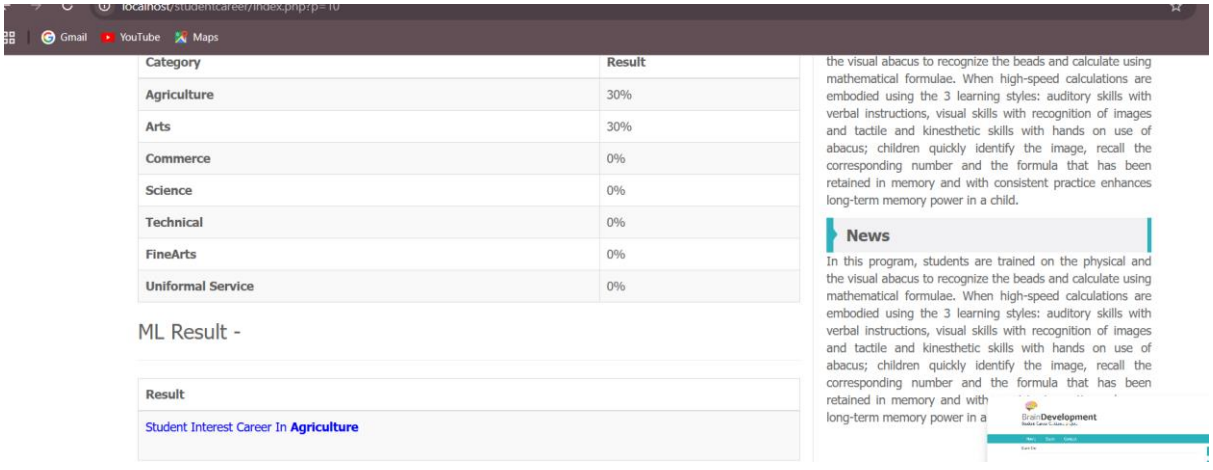
SNAPSHOT17. Student Is Solving the Questions



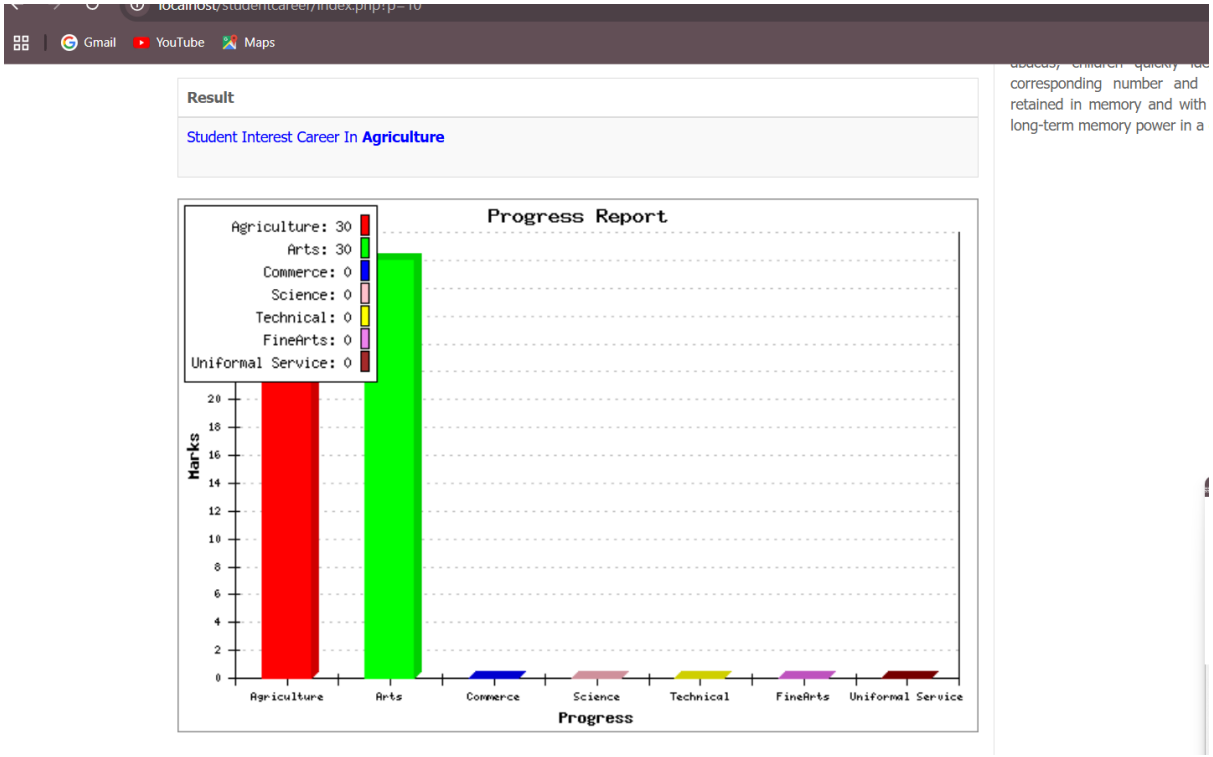
SNAPSHOT18. Student Has End the Exam



SNAPSHOT19. Student Result



SNAPSHOT20. Student Interest in which Field Predicted



SNAPSHOT21. Student Interest in which Field Predicted and Graph is shown

ADVANTAGES

1. **Gives Personal Career Advice:**
   It helps students know which career suits them best based on their answers in a simple MCQ test.
2. **Fair and Accurate:**
   The system uses smart technology (Random Forest) to make decisions based on data, not guesses or opinions.
3. **Understands Student Strengths:**
   By analysing answers, it finds out what subjects or skills a student is good at.
4. **Easy to Use**:
   Students just take a multiple-choice test, and the system does the rest—no need for long interviews or paperwork.
5. **Works for Everyone:**
   It can be used in any school or college and helps many students at the same time without needing a human counsellor for each one.

APPLICATION

1. **School and College Career Guidance:**
   Schools can use it to help students choose the right career stream (like science, commerce, or arts) based on their interests and strengths.
2. **Career Counselling Centres:**
   Counsellors can use the system to give faster and more accurate advice by using student test results instead of only manual methods.
3. **Online Career Guidance Platforms:**
   Websites and apps can use this system to offer instant career suggestions to students after they complete a short MCQ test.
4. **Entrance Exam Preparation Help:**
   Based on the student's answers, it can suggest which competitive exams or career paths (like engineering, medicine, design, etc.) are a good fit.
5. **Self-Assessment Tool for Students:**
   Students can take the test on their own and get ideas about careers that match their skills—even without a teacher or counsellor.

CONCLUSION

The integration of a student career recommendation system tailored to student preferences holds significant promise as a transformative tool in the realm of academic and professional guidance. By systematically analysing a wide range of individual attributes—such as personal interests, innate skills, academic achievements, and extracurricular experiences—the system can generate career suggestions that are closely aligned with the student's unique profile. This personalized approach enables the identification of career opportunities that resonate with the student's values, goals, and long-term aspirations, thereby reducing uncertainty and enhancing confidence in the decision-making process. In an educational landscape where students are often overwhelmed by the sheer volume of career choices, such a system can act as a compass, offering direction based on evidence rather than guesswork.

Moreover, the incorporation of advanced technologies, particularly artificial intelligence and machine learning algorithms, significantly enhances the accuracy, efficiency, and adaptability of career recommendation systems. These intelligent algorithms are capable of processing large volumes of complex data to detect meaningful patterns and correlations that may not be immediately visible through traditional guidance methods. As a result, the recommendations generated are not only more nuanced and context-aware, but also continuously improve over time through adaptive learning mechanisms. This technological advantage empowers students with high-quality, data-driven insights that are customized to their evolving preferences and educational progress, ultimately facilitating more strategic, informed, and confident choices about their future career paths.

FUTURE SCOPE

1. **Better Accuracy**: The app can become more accurate in predicting careers as it learns from more student data over time.
2. **Real-Time Updates**: It can use real-time data from schools, online courses, and tests to give updated career suggestions.
3. **Personalized Guidance**: The app can provide career paths based on each student's unique interests, strengths, and goals.
4. **Industry Trends**: Future versions could include current job market trends to suggest careers that are in demand.
5. **More Student Information**: It can include more factors like hobbies, soft skills, or personality traits to improve predictions.
6. **Smarter Learning**: The Random Forest model can be trained continuously to improve itself with new data and better results.
7. **Available Anywhere**: The app can be made available online or in the cloud so students from any location can use it.
8. **Support for All Languages**: It could support multiple languages so students from different regions can easily understand and use it.
9. **Easy to Use**: With a simple and friendly design, even students with little tech knowledge can use the app comfortably.
10. **Feedback System**: Students can give feedback, and the app can learn from that to improve future suggestions.
11. **Voice and Chat Support**: It may include AI chatbots or voice assistants to help students ask questions and get career advice instantly.
12. **Safe and Private**: Future apps can be designed to protect student data and ensure everything stays private.
13. **Works with Schools**: The app could be linked with schools or colleges to provide career support directly through the education system.
14. **Career Growth Tracking**: It might help students track their progress and prepare step-by-step for their dream job.
15. **Accessible for Everyone**: The app can be designed to work well for students with disabilities or learning challenges too.

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CONTRIBUTION OF THE PROJECT

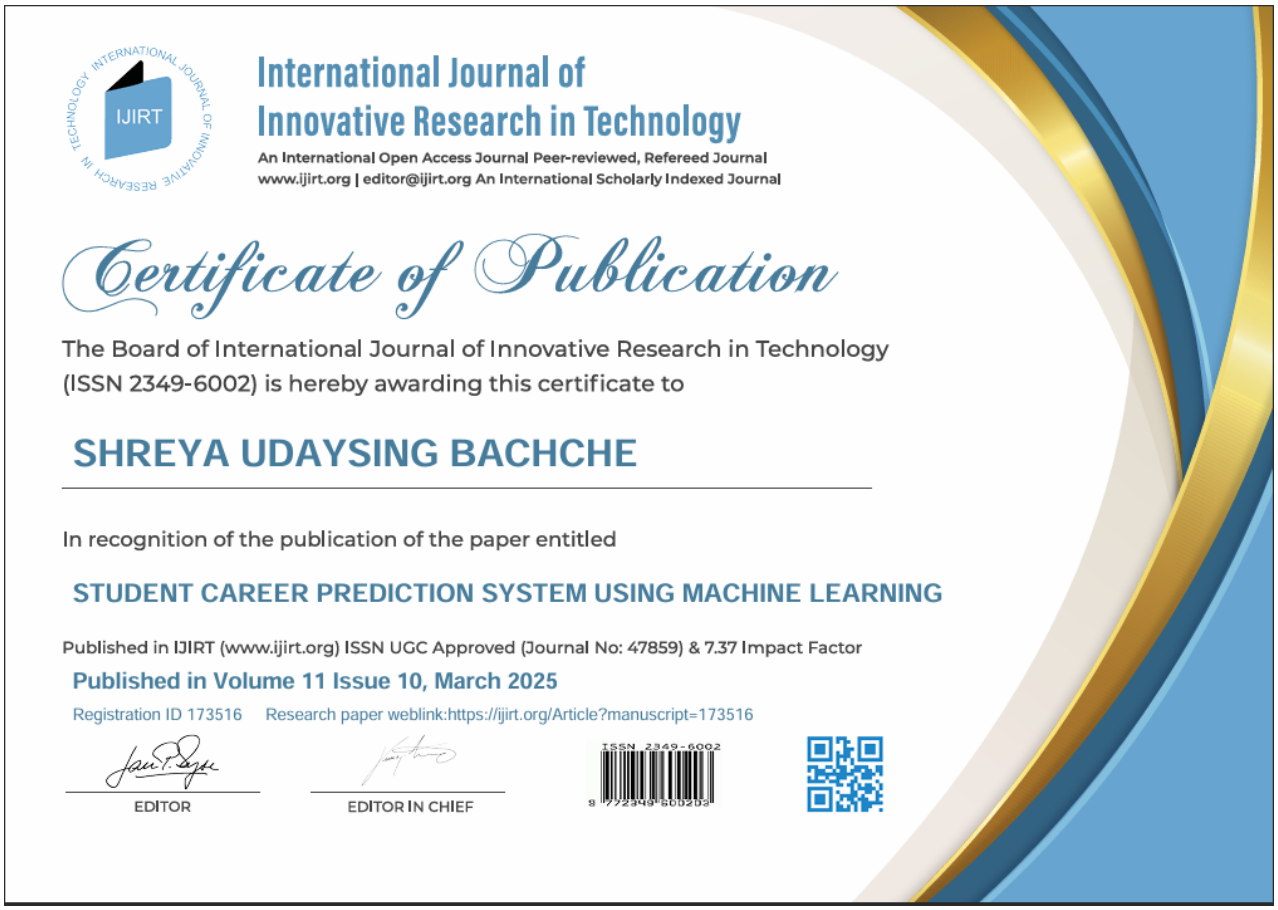
We Have Published Research Paper on IJIRT Website

Paper Title: Student Career Prediction using Machine Learning.

Paper ID : 173516

Link : https://ijirt.org/Article?manuscript=173516

CERTIFICATE



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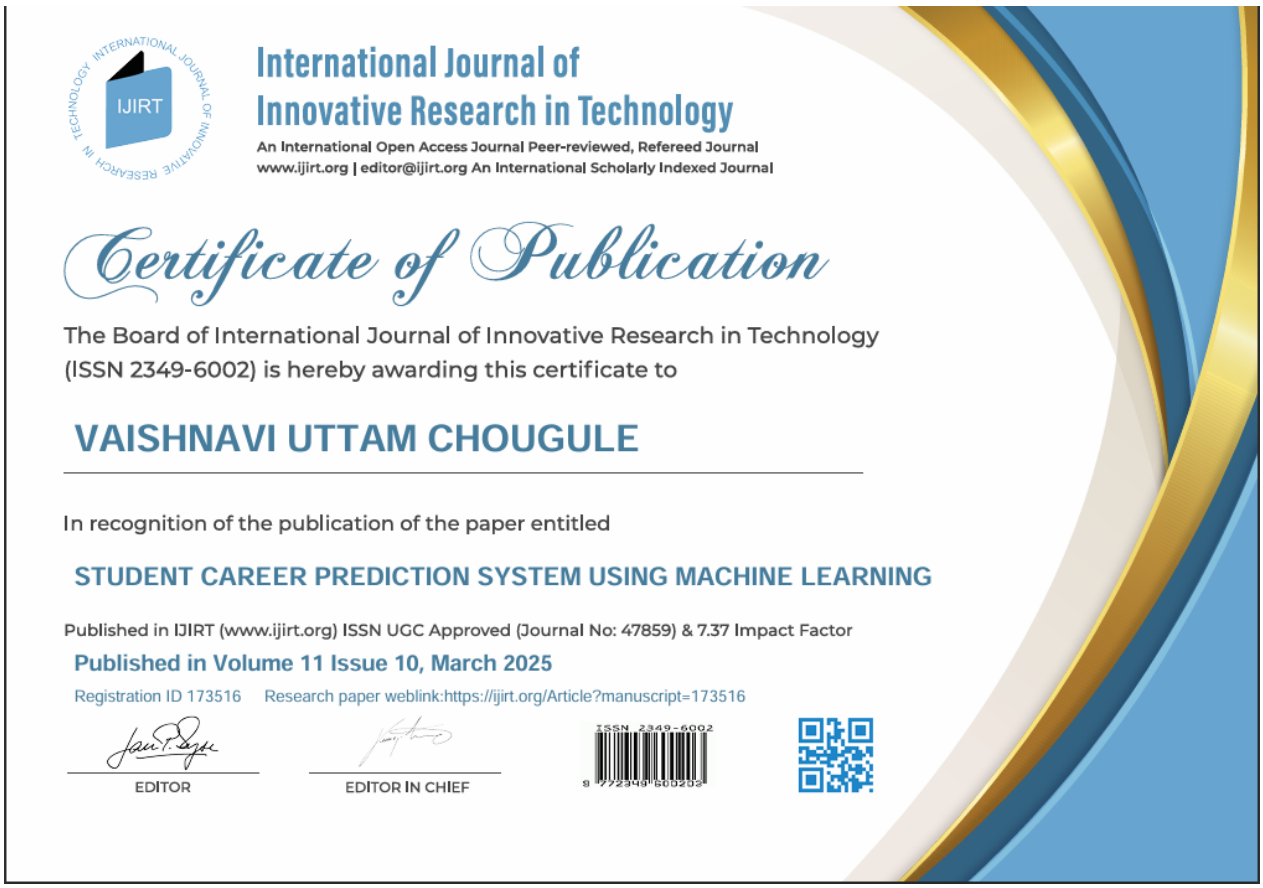
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